A First Course in Abstract Algebra

Section 27: Prime and Maximal Ideals

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

Contents

27 Prime and Maximal Ideals 3

Exercise 27.1: Prime and maximal ideals in a six-element ring 3

Exercise 27.2: Prime and maximal ideals in a twelve-element ring 3

Exercise 27.3: A product of two two-element fields 3

Exercise 27.4: A product involving a four-element ring 3

Exercise 27.5: Quadratic quotients over the three-element field 3

Exercise 27.6: A cubic family with fixed quadratic term 3

Exercise 27.7: A cubic family with a variable coefficient 3

Exercise 27.8: Quadratics with a linear term over five elements 4

Exercise 27.9: Quadratics with constant term one 4

Exercise 27.10: Correcting maximal ideal 4

Exercise 27.11: Correcting prime ideal 4

Exercise 27.12: Prime field 4

Exercise 27.13: Principal ideal 4

Exercise 27.14: True and false statements 4

Exercise 27.15: A maximal ideal of an integer product 5

Exercise 27.16: A prime ideal that is not maximal 5
Exercise 27.17: An ideal that is not prime 5
Exercise 27.18: A reducible rational quadratic quotient 5
Exercise 27.19: An irreducible rational quadratic quotient 5
Exercise 27.20: Synopsis: maximality produces a field quotient 5
Exercise 27.21: Synopsis: a field quotient forces maximality 6
Exercise 27.22: Synopsis: ideals in a polynomial ring 6
Exercise 27.23: Synopsis: maximal polynomial ideals are irreducible 6
Exercise 27.24: Finite prime quotients 6
Exercise 27.25: Different cyclic-ring subrings 6
Exercise 27.26: Two different prime-field subrings 6
Exercise 27.27: The domain case 6
Exercise 27.28: Direct proof that maximal ideals are prime 7
Exercise 27.29: Maximal ideals and simple quotient rings 7
Exercise 27.30: Nonzero prime ideals of a one-variable polynomial ring 7
Exercise 27.31: Divisibility and membership 7
Exercise 27.32: The ideal generated by two polynomials 7
Exercise 27.33: A one-variable Nullstellensatz 7
Exercise 27.34: The sum of two ideals 8
Exercise 27.35: The product of two ideals 8
Exercise 27.36: The ideal quotient 8
Exercise 27.37: A one-sided matrix ideal 8
Exercise 27.38: Simplicity of a matrix ring 8

27 Prime and Maximal Ideals

Exercise 27.1: Prime and maximal ideals in a six-element ring

Every ideal is $\langle d\rangle$ for $d \mid 6$, and $\mathbb{Z}_6/\langle d\rangle \cong \mathbb{Z}_d$. A proper ideal is prime exactly when its quotient is an integral domain, and maximal exactly when its quotient is a field. Among the divisors $1, 2, 3, 6$, exactly $2, 3$ are prime. Thus both lists are $\langle 2\rangle = \{0, 2, 4\}$ and $\langle 3\rangle = \{0, 3\}$.

Exercise 27.2: Prime and maximal ideals in a twelve-element ring

For $d \mid 12$, reduction modulo d gives $\mathbb{Z}_{12}/\langle d\rangle \cong \mathbb{Z}_d$. The quotient is a domain, equivalently a field in this finite case, exactly when $d = 2$ or 3. Hence the prime ideals and the maximal ideals are both $\langle 2\rangle = \{0, 2, 4, 6, 8, 10\}$ and $\langle 3\rangle = \{0, 3, 6, 9\}$.

Exercise 27.3: A product of two two-element fields

In a product $A \times B$ with identities, every ideal is $I \times J$: multiply (a, b) by $(1, 0)$ and $(0, 1)$ to separate the coordinates. A domain quotient cannot retain both nonzero factors, since $(1, 0)(0, 1) = 0$. Thus the prime and maximal ideals of $\mathbb{Z}_2 \times \mathbb{Z}_2$ are exactly $\{0\} \times \mathbb{Z}_2$ and $\mathbb{Z}_2 \times \{0\}$. Each quotient is $\mathbb{Z}_2$.

Exercise 27.4: A product involving a four-element ring

The same product argument shows that a prime ideal must be all of one factor times a prime ideal in the other. In $\mathbb{Z}_2$ the only prime ideal is zero, and in $\mathbb{Z}_4$ it is $\langle 2\rangle = \{0, 2\}$. Therefore the only prime ideals are

$$\{0\} \times \mathbb{Z}_4, \qquad \mathbb{Z}_2 \times \{0, 2\}.$$

Both quotients are $\mathbb{Z}_2$, so these are also exactly the maximal ideals.

Exercise 27.5: Quadratic quotients over the three-element field

A quotient by a nonconstant polynomial is a field exactly when that polynomial is irreducible. A quadratic over a field is irreducible exactly when it has no root. In $\mathbb{Z}_3$, the possible squares are $0, 1$. The equation $x^2 = -c$ has no solution exactly when $-c = 2$, or $c = 1$. Thus only $c = 1$ works.

Exercise 27.6: A cubic family with fixed quadratic term

A cubic over a field is reducible exactly when it has a root. At $x = 0, 1, 2$, the values of $x^3 + x^2$ modulo three are $0, 2, 0$. Thus $x^3 + x^2 + c$ has a root if $c = 0$ or 1, but for $c = 2$ its values are $2, 1, 2$, all nonzero. The required value is $c = 2$.

Exercise 27.7: A cubic family with a variable coefficient

Evaluate $x^3 + cx^2 + 1$ at $0, 1, 2$. For $c = 0$ the value at 2 is zero; for $c = 1$ the value at 1 is zero; for $c = 2$ the three values are $1, 1, 2$, all nonzero. Thus exactly $c = 2$ makes the cubic irreducible and the quotient a field.

Exercise 27.8: Quadratics with a linear term over five elements

Since the characteristic is not two, $x^2 + x + c$ has a root exactly when the discriminant $1 - 4c$ is a square. The squares in $\mathbb{Z}_5$ are $0, 1, 4$. For $c = 0, 1, 2, 3, 4$, the discriminants are respectively $1, 2, 3, 4, 0$. Thus the nonsquare cases are $c = 1, 2$, and precisely those quotients are fields.

Exercise 27.9: Quadratics with constant term one

For $x^2 + cx + 1$, the discriminant is $c^2 - 4$. As c runs through $0, 1, 2, 3, 4$ in $\mathbb{Z}_5$, the discriminants are $1, 2, 0, 0, 2$. Nonsquares occur for $c = 1, 4$, so exactly those values give fields.

Exercise 27.10: Correcting maximal ideal

A maximal ideal M is a *proper* ideal of R such that the only ideals containing it are M and R. Merely saying that no other ideal contains it would make R the only candidate, because R contains every ideal.

Exercise 27.11: Correcting prime ideal

In a commutative ring, a prime ideal P is a proper ideal such that $ab \in P$ implies $a \in P$ or $b \in P$ for all $a, b \in R$. It need not be generated by a prime number; for example $(x) \subset \mathbb{Q}[x]$ is prime because its quotient is $\mathbb{Q}$.

Exercise 27.12: Prime field

The definition is correct: a prime field has no proper subfields. Equivalently it equals the subfield generated by its unity. Up to isomorphism these fields are $\mathbb{Q}$ and $\mathbb{Z}_p$ for primes p.

Exercise 27.13: Principal ideal

The definition is correct. In a commutative ring with unity, the smallest ideal containing a is $(a) = aR = \{ar : r \in R\}$. It contains $a = a1$, is closed under differences and absorption, and every ideal containing a contains all its multiples.

Exercise 27.14: True and false statements

(a) False. $(0) \subset \mathbb{Z}$ is prime but not maximal, since $\mathbb{Z}$ is a domain and not a field.

(b) True. The quotient by a maximal ideal is a field and hence a domain.

(c) True. Every subfield of $\mathbb{Q}$ contains 1, all integers, and all nonzero integer inverses, hence all rationals.

(d) False. The prime subfield of $\mathbb{C}$ is $\mathbb{Q}$.

(e) True. The unity generates a copy of $\mathbb{Q}$ in characteristic zero or of $\mathbb{Z}_p$ in characteristic p.

(f) True. $\mathbb{Z}_2 \times \mathbb{Z}_2$ has zero divisors and contains the diagonal field $\{(0,0),(1,1)\}$.

(g) True. Send m/n to $(m1)(n1)^{-1}$; characteristic zero ensures no nonzero integer maps to zero.

(h) False. $(x^2) \subset F[x]$ is not prime: x^2 belongs but x does not.

(i) True. Choose a nonzero element of smallest degree and use the division algorithm; include the zero ideal separately.

(j) False. (x^2) is principal but its quotient has a nonzero nilpotent element $x + (x^2)$, so is not a field.

Exercise 27.15: A maximal ideal of an integer product

Let $M = 2\mathbb{Z} \times \mathbb{Z}$. Projection onto the first coordinate followed by reduction modulo two is onto $\mathbb{Z}_2$ and has kernel M. Thus $(\mathbb{Z} \times \mathbb{Z})/M \cong \mathbb{Z}_2$, a field, and M is maximal.

Exercise 27.16: A prime ideal that is not maximal

Let $P = \{0\} \times \mathbb{Z}$. Projection onto the first coordinate gives quotient $\mathbb{Z}$, a domain but not a field. Thus P is prime and not maximal. Explicitly $P \subsetneq 2\mathbb{Z} \times \mathbb{Z} \subsetneq \mathbb{Z} \times \mathbb{Z}$.

Exercise 27.17: An ideal that is not prime

Let $I = 4\mathbb{Z} \times \mathbb{Z}$. It is proper and nonzero. Neither factor $(2,0)$ of $(2,0)(2,0) = (4,0)$ lies in I, but their product does. Hence I is not prime.

Exercise 27.18: A reducible rational quadratic quotient

Since $x^2 - 5x + 6 = (x-2)(x-3)$, the defining polynomial is reducible over $\mathbb{Q}$. In the quotient, the two linear-factor cosets are nonzero (their degrees are too small to be multiples of the quadratic), and their product is zero. Thus the quotient is not a field.

Exercise 27.19: An irreducible rational quadratic quotient

The discriminant of $x^2 - 6x + 6$ is $36 - 24 = 12$, whose square root $2\sqrt{3}$ is irrational. Thus the quadratic has no rational root and is irreducible over $\mathbb{Q}$. Its generated ideal is maximal, so the quotient is a field, isomorphic to $\mathbb{Q}(\sqrt{3})$ by evaluating at $3 + \sqrt{3}$.

Exercise 27.20: Synopsis: maximality produces a field quotient

If M is maximal and $a \notin M$, the ideal $M + (a)$ must be R, so $1 = m + ra$ for some $m \in M, r \in R$. Passing to cosets gives $(r + M)(a + M) = 1 + M$, so every nonzero coset has an inverse and R/M is a field.

Exercise 27.21: Synopsis: a field quotient forces maximality

If R/M is a field and I is an ideal properly containing M, choose $a \in I \setminus M$; an inverse of $a + M$ gives $ra - 1 \in M$ for some $r \in R$, so $1 \in I$ and $I = R$. Thus M has no proper intermediate ideal and is maximal.

Exercise 27.22: Synopsis: ideals in a polynomial ring

For a nonzero ideal $I \subset F[x]$, choose $g \in I$ of smallest degree. Dividing any $f \in I$ by g gives a remainder in I of smaller degree, hence zero; therefore $I = (g)$. The zero ideal is (0).

Exercise 27.23: Synopsis: maximal polynomial ideals are irreducible

If (p) is maximal but $p = fg$ with both factors of positive degree, then $(p) \subsetneq (f) \subsetneq F[x]$, a contradiction. Hence a nonconstant generator of a maximal ideal must be irreducible.

Exercise 27.24: Finite prime quotients

If P is prime, R/P is an integral domain and is finite because R is finite. For any nonzero a in a finite domain, the multiplication map $x \mapsto ax$ is injective by cancellation, hence surjective. It therefore takes some x to 1, giving an inverse for a. Thus R/P is a field, and P is maximal.

Exercise 27.25: Different cyclic-ring subrings

It is possible when the subrings need not share the ambient unity. In $R = \mathbb{Z}_2 \times \mathbb{Z}_3$, the subrings $\mathbb{Z}_2 \times \{0\}$ and $\{0\} \times \mathbb{Z}_3$ are isomorphic to $\mathbb{Z}_2$ and $\mathbb{Z}_3$, respectively. Their identities are $(1, 0)$ and $(0, 1)$, not the ambient $(1, 1)$. This distinction avoids incorrectly equating their characteristics with that of R.

Exercise 27.26: Two different prime-field subrings

The same example $\mathbb{Z}_p \times \mathbb{Z}_q$, for distinct primes p, q, contains the field subrings $\mathbb{Z}_p \times \{0\}$ and $\{0\} \times \mathbb{Z}_q$. Each is a field under the inherited operations, with its own identity. Thus the answer is yes under the book's subring convention.

Exercise 27.27: The domain case

In an integral domain, a nonzero idempotent e satisfies $e(e - 1) = 0$, so $e = 1$. Therefore the identity of any field subring is the ambient identity. A copy of $\mathbb{Z}_p$ forces the ambient characteristic to be p, and a copy of $\mathbb{Z}_q$ forces it to be q. Since $p \neq q$ is impossible, no such domain exists. Equivalently distinct nonzero orthogonal identities would create zero divisors.

Exercise 27.28: Direct proof that maximal ideals are prime

Suppose $ab \in M$ and $a \notin M$. The ideal $M + (a)$ properly contains M, so maximality makes it R. Hence $1 = ra + m$ for some $r \in R, m \in M$. Multiply by b:

$$b = rab + mb \in M,$$

since both terms belong to M. This proves $a \in M$ or $b \in M$, and M is proper by definition, so it is prime.

Exercise 27.29: Maximal ideals and simple quotient rings

Ideals J containing N correspond to ideals J/N of R/N: inverse image under the quotient map reverses this assignment. To see the image direction, multiplication by $r + N$ sends $j + N$ to $rj + N \in J/N$; the inverse-image direction follows from the ideal test. Therefore the only ideals of R/N are zero and the whole quotient exactly when the only ideals containing N are N and R. Requiring a nontrivial quotient is equivalent to $N \neq R$. These statements say precisely that N is maximal if and only if R/N is simple.

Exercise 27.30: Nonzero prime ideals of a one-variable polynomial ring

Write the proper nonzero prime ideal as (p), using the principal-ideal theorem for $F[x]$. If $p = fg$ with $\deg f, \deg g > 0$, primality puts f or g in (p), impossible by degree. Thus p is irreducible. The quotient by an irreducible polynomial over a field is a field, so (p) is maximal.

Exercise 27.31: Divisibility and membership

By definition, $(f) = \{hf : h \in F[x]\}$. Thus $g \in (f)$ means exactly that $g = hf$ for some polynomial h, which is the statement $f \mid g$. This includes $f = 0$: membership and divisibility both then require $g = 0$.

Exercise 27.32: The ideal generated by two polynomials

The set $N = (f, g)$ contains zero and is closed under subtraction:

$$(rf + sg) - (r'f + s'g) = (r - r')f + (s - s')g.$$

Multiplying by any $h \in F[x]$ gives $(hr)f + (hs)g$, again in N, so it is an ideal. If both f, g were irreducible and had distinct degrees, their only common divisors would be constants: a nonconstant common divisor of an irreducible is an associate of it, forcing equal degrees. The Euclidean algorithm then gives $uf + vg = 1$, which would make $N = F[x]$. This contradicts the stated hypothesis; thus they cannot both be irreducible.

Exercise 27.33: A one-variable Nullstellensatz

Plan. Replace the finitely generated ideal by a single greatest common divisor. Assume the Fundamental Theorem of Algebra. Write $(f_1, \ldots, f_r) = (h)$ using the principal-ideal theorem. If

h is a nonzero constant, the ideal is the whole ring and the assertion is immediate. If $h = 0$, all f_i vanish everywhere; then g vanishes everywhere, so it is the zero polynomial and belongs to the ideal. Otherwise factor $h = c\prod_j (x - a_j)^{e_j}$. Its zeros are exactly the common zeros of the f_i, since h divides every f_i and is their polynomial linear combination. Each a_j is therefore a zero of g. Taking $m \geq \max_j e_j$ makes $h \mid g^m$, so $g^m \in (f_1, \ldots, f_r)$. Conversely assume the stated Nullstellensatz. If a nonconstant f had no complex zero, the condition on $g = 1$ would be vacuously true. It would follow that $1 = 1^m \in (f)$, impossible since a nonconstant polynomial cannot divide 1. Thus every nonconstant complex polynomial has a zero.

Exercise 27.34: The sum of two ideals

(a) Zero is $0 + 0 \in A + B$. Subtracting $a + b$ and $a' + b'$ yields $(a - a') + (b - b') \in A + B$. For $r \in R$, both $r(a + b) = ra + rb$ and $(a + b)r = ar + br$ lie in $A + B$. Thus this is an ideal even if R is noncommutative. **(b)** Since $0 \in A \cap B$, write $a = a + 0$ and $b = 0 + b$. This proves $A \subset A + B$ and $B \subset A + B$.

Exercise 27.35: The product of two ideals

(a) The zero sum can be represented as $0 \cdot 0$. Sums of finite sums of products remain such sums; a negative is obtained by replacing a_i by $-a_i$. Also

$$r\sum_i a_i b_i = \sum_i (ra_i)b_i, \qquad (\sum_i a_i b_i)r = \sum_i a_i(b_i r),$$

and the changed factors stay in the appropriate ideals. Hence AB is an ideal. **(b)** Each $a_i b_i$ lies in A by right absorption and in B by left absorption. Both ideals are closed under finite sums, so $AB \subset A \cap B$.

Exercise 27.36: The ideal quotient

Let $I = A : B = \{r : rb \in A \text{ for every } b \in B\}$. Zero belongs to I. If $r, s \in I$, then $(r-s)b = rb-sb \in A$ for every $b \in B$, so $r - s \in I$. If $t \in R$, then $(tr)b = t(rb) \in A$. Thus $tr \in I$, and commutativity gives the other side too. Therefore $A : B$ is an ideal.

Exercise 27.37: A one-sided matrix ideal

Let S be the matrices with second row zero. Differences and products of such matrices still have second row zero, so S is a subring. For any $B \in M_2(F)$, the second row of AB is zero whenever $A \in S$, so S is a right ideal. Using matrix units, $E_{11} \in S$ but $E_{21}E_{11} = E_{21} \notin S$. Therefore multiplication on the left by an arbitrary ambient matrix need not stay in S, and S is not a left ideal.

Exercise 27.38: Simplicity of a matrix ring

Let I be a nonzero two-sided ideal in $M_2(\mathbb{Z}_2)$, and choose $A \in I$ with nonzero entry $a_{ij} = 1$. For any $r, s \in \{1, 2\}$,

$$E_{ri}AE_{js} = a_{ij}E_{rs} = E_{rs} \in I.$$

Thus I contains all four matrix units. Every binary matrix is the sum of the matrix units corresponding to its nonzero entries, so $I = M_2(\mathbb{Z}_2)$. Hence the only two-sided ideals are zero and the whole ring, proving simplicity.